ANDREW NEIL

AI ENGINEERING • AGENTIC SYSTEMS • PYTHON BACKEND

Mumbai, India • +91 81086 85955 • **andrewneil1912@gmail.com**
andrewneil.dev • LinkedIn • GitHub

AI & Python Backend Engineer building fault-tolerant multi-agent workflows, desktop/browser automation pipelines, and production Django backends. Proven track record shipping ERP backend systems, delivering 3 international freelance AI/automation contracts (5-star rated), and engineering deterministic LLM execution harnesses backed by 336 automated regression tests and 3 published IJIRT papers.

TECHNICAL SKILLS

Domain	Technologies & Engineering Capabilities
Agentic AI & LLMs	LLM Task Decomposition, Tool Calling, Structured JSON Actions, Persistent Workflow State, LangChain, Gemini / OpenAI / Anthropic APIs, OmniRoute Multi-Model Routing, WebLLM (Local Qwen2.5 Inference), RAG & Vector Search, Token Tracking
Backend & Full-Stack	Python, Django, FastAPI, REST APIs, JavaScript, React 19, Vite, Tailwind CSS v4, Flutter/Dart, JWT Auth, Git/GitHub, Linux/CLI, SMTP Pipelines
Reliability & Automation	pytest (336+ Regression Suite), Schema & Payload Validation, Fresh-Observation Loops, Outcome Verification, Bounded Retries, Playwright, Selenium, PyAutoGUI
Data, Vision & IoT	Pandas, NumPy, Parquet, pandas-ta, YOLOv8, OpenCV, OCR/ANPR, scikit-learn, PyTorch (LSTM), Walk-Forward Backtesting, ESP32-S3, NFC/RFID, Bluetooth LE

PROFESSIONAL EXPERIENCE

Backend Engineer Intern — 4iStudio | Python, Django, REST APIs, Relational DB **Aug 2026 – Present**

- Architect and maintain production Django backend workflows for a multi-role polytechnic ERP managing student lifecycle records, attendance tracking, and institutional marks entry.
- Engineered automated marksheet and certificate generation pipelines with role-based access control across academic departments.

Freelance Python & AI Developer — Upwork | LangChain, OpenAI/Claude APIs, Selenium **2023 – 2025 • 3 Contracts (2 Five-Star Reviews)**

- Built a stateful LangChain CLI integrating OpenAI and Anthropic Claude APIs to generate multi-variant responses, support iterative regeneration, and persist selected outputs as dynamic context.
- Delivered an automated website-availability monitoring system with real-time SMTP alerting and Selenium browser scraping pipelines for international clients.

FLAGSHIP ENGINEERING PROJECTS

OSCAR / OSCAR 2.0 — Multi-Agent AI Desktop & Browser Assistant | Python, Flutter, Gemini/OpenAI/Claude, OmniRoute, Playwright, pytest **336 Passing Tests**

- Architected a multi-agent orchestration engine converting natural-language intents into multi-step workflows across browser (Playwright), desktop UI (PyAutoGUI/OpenCV), and terminal environments.
- Implemented vision-driven observation-action loops dispatching structured JSON actions with persistent workflow state, record/replay, OmniRoute multi-provider routing, and token/cost telemetry.
- Engineered a fault-tolerant execution harness with strict tool payload validation, outcome verification checkpoints, bounded retries, duplicate-input guards, and a 336-test automated `pytest` suite.

TradeAssist — Quantitative Market Pipeline & Backtesting Engine | Python, SmartAPI, Pandas, Parquet, PyTorch **13 Walk-Forward Folds**

- Built a live market-data ingestion pipeline and historical equity backtesting engine modeling technical signals, position sizing, 5-bps slippage, transaction costs, and trailing-stop execution.
- Evaluated strategy variants across 13 walk-forward folds and PyTorch LSTM experiments, achieving +24.17% return, 6.75% CAGR, and -11.03% max drawdown across 86 trades on the IRFC SMA 20/50 crossover strategy.

FinVerse 2.0 — AI & IoT B2B Payment Platform | Django, React 19, Flutter, ESP32-S3, NFC/RFID **Active Development**

- Developing a B2B payment and M2M commerce platform combining an AI financial copilot, automated fraud detection, and hardware-backed ESP32-S3 NFC terminal authentication.

Dynamic Toll Charging System | Python, YOLOv8, OpenCV, OCR/ANPR, RFID **1st Place Winner • IJIRT Paper**

- Won 1st place for an automated distance-based tolling and wallet pipeline correlating YOLOv8 vehicle detection, ANPR license-plate OCR, and RFID entry/exit verification.

KYC Smart Card — Identity & Document Platform | Django, Flutter, ESP32, RFID, Bluetooth LE, JWT **2 IJIRT Papers (2025, 2024)**

- Co-developed a DigiLocker-inspired verification system featuring dual-mode RFID smartcard and ESP32 Bluetooth LE authentication, JWT session security, and encrypted document workflows.

EDUCATION & PEER-REVIEWED PUBLICATIONS

B.E. in Computer Engineering — Xavier Institute of Engineering (XIE), Mumbai **Expected 2027 • CGPA: 7.48/10**

- **IJIRT Publications:** *Dynamic Toll Charging System* (Vol. 12, Issue 8, 2026); *KYC Smart Card Dual-Mode Authentication* (Vol. 12, Issue 1, 2025); *KYC Smart Card RFID Repository* (Vol. 11, Issue 7, 2024).